

A_sub commutator closure step 2 — [$\hat{T}_{\text{tick}}$, $\hat{H}_{\text{sub}}$] substrate discreteness signature

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A_sub commutator closure step 2 — [$\hat{T}_{\text{tick}}$, $\hat{H}_{\text{sub}}$]: the substrate's discreteness signature

1. Setup — $\hat{T}_{\text{tick}}$ and $\hat{H}_{\text{sub}}$ on Wallach K-types

1.1 $\hat{T}_{\text{tick}}$ — substrate-tick transition operator

Per Task #322 v0.2 Section 10 (Sunday 2026-05-24): the substrate-tick transition operator $\hat{T}_{\text{tick}}$ advances substrate state by one Koons tick $t_K \approx 10^{(-120)}$ s.

$\hat{T}_{\text{tick}}$ is unitary (substrate evolution preserves Hilbert space norm) and discrete (operates per Koons tick, not as continuous Hamiltonian flow).

v0.1 simplest model (per Sunday v0.2): *$\hat{T}_{\text{tick}}$ advances K-type index by 1 step in a substrate-natural ordering:* $\hat{T}_{\text{tick}} V(m_1, m_2) = V(m_1, m_2 + 1)$

This is the SIMPLEST candidate model. The full K59 cyclotomic 7-step mechanism on GF(128) gives a more complex transition pattern that refines this in v0.2+ (multi-week work).

(Honest scope: this simplest model may not capture full $\hat{T}_{\text{tick}}$ structure. Cal #27 STANDING reflexive: feels too simple to be the full mechanism; flagged as v0.1 candidate; full mechanism via K59 cyclotomic chain expected to refine.)

1.2 $\hat{H}_{\text{sub}}$ — substrate Hamiltonian (Casimir multiplication)

Per Task #322 v0.1 Section 2.2 + T2441 STRUCTURALLY VERIFIED: the substrate Hamiltonian $\hat{H}_{\text{sub}}$ acts diagonally on Wallach K-types via Casimir eigenvalue:

$$\hat{H}_{\text{sub}} V(m_1, m_2) = C_2(K) \cdot V(m_1, m_2)$$

where $C_2(K)$ is the Casimir-2 eigenvalue of the K-type, given by: $C_2(K) = m_1(m_1 + n_C - 1) + m_2(m_2 + N_c - 2) = m_1(m_1 + 4) + m_2(m_2 + 1)$ (with $n_C = 5$, $N_c = 3$ BST primaries)

1.3 $\Delta C_2(K)$ per tick

Per the simplest model $\hat{T}_{\text{tick}} : (m_1, m_2) \rightarrow (m_1, m_2 + 1)$:

$$\begin{aligned} \Delta C_2(K) &:= C_2(m_1, m_2 + 1) - C_2(m_1, m_2) = (m_2 + 1)((m_2 + 1) + N_c - 2) - m_2(m_2 + N_c - 2) \\ &= (m_2 + 1)(m_2 + 1) - m_2 m_2 + (N_c - 2)((m_2 + 1) - m_2) \\ &= 2 m_2 + 1 + N_c - 2 = 2 m_2 + N_c - 1 = 2 m_2 + 2 \text{ (with } N_c = 3) \end{aligned}$$

2. The commutator computation

2.1 Direct calculation

Apply $\hat{T}_{\text{tick}} \circ \hat{H}_{\text{sub}}$: $\hat{T}_{\text{tick}} \circ \hat{H}_{\text{sub}} V_K = \hat{T}_{\text{tick}} \cdot C_2(K) \cdot V_K = C_2(K) \cdot V_{\{K+1 \text{ (in } m_2)\}}} = C_2(m_1, m_2) \cdot V(m_1, m_2 + 1)$

Apply $\hat{H}_{\text{sub}} \circ \hat{T}_{\text{tick}}$: $\hat{H}_{\text{sub}} \circ \hat{T}_{\text{tick}} V_K = \hat{H}_{\text{sub}} V(m_1, m_2 + 1) = C_2(m_1, m_2 + 1) \cdot V(m_1, m_2 + 1)$

Commutator: $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] V_K = \hat{T}_{\text{tick}} \hat{H}_{\text{sub}} V_K - \hat{H}_{\text{sub}} \hat{T}_{\text{tick}} V_K = (C_2(m_1, m_2) - C_2(m_1, m_2 + 1)) \cdot V(m_1, m_2 + 1) = -\Delta C_2(K) \cdot V(m_1, m_2 + 1) = -(2 m_2 + N_c - 1) \cdot \hat{T}_{\text{tick}} V_K = -(2 m_2 + 2) \cdot \hat{T}_{\text{tick}} V_K$

2.2 The structural identity

$$**[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] = -(2 \hat{Q} + N_c - 1) \cdot \hat{T}_{\text{tick}} = -(2 \hat{Q} + 2) \cdot \hat{T}_{\text{tick}} \text{ on } H^2(D_{IV}^5).**$$

(Using $\hat{Q} V_K = m_2 \cdot V_K$, so $2 \hat{Q} + N_c - 1$ acts diagonally with eigenvalue $2 m_2 + N_c - 1 = 2 m_2 + 2$.)

3. Substrate-mechanism interpretation

3.1 The substrate's discreteness signature

The commutator $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ is NON-ZERO at finite t_K . This is structurally important:

- If substrate evolution were continuous Hamiltonian flow $\hat{T}_{\text{tick}} = \exp(-i \hat{H}_{\text{sub}} t_K / \hbar)$, then $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] = 0$ identically (any function of $\hat{H}$ commutes with $\hat{H}$).
- The non-vanishing of $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ establishes that substrate evolution IS discrete and NOT a continuous Hamiltonian flow.
- The Hamiltonian $\hat{H}_{\text{sub}}$ is the coarse-grained continuum limit of the discrete $\hat{T}_{\text{tick}}$; in the limit $t_K \rightarrow 0$ with $N \cdot t_K \rightarrow t$ fixed, $\hat{T}_{\text{tick}}^N \rightarrow \exp(-i \hat{H}_{\text{sub}} t / \hbar)$ and the commutator vanishes.
- For finite t_K (which IS the substrate's actual regime), the commutator captures the substrate's discreteness.

Substrate discreteness signature: $||[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]|| / ||\hat{H}_{\text{sub}}||$ measures the substrate's deviation from continuous Hamiltonian flow.

3.2 Charge-weighted discreteness

The commutator scales with $(2\hat{Q} + N_c - 1)$: - Higher-charge K-types have larger discreteness — charge m_2 amplifies the substrate's per-tick non-commutativity - Neutral K-types ($m_2 = 0$) still have commutator $-(N_c - 1) = -2$ — non-zero from N_c contribution - Substrate ground state $V_-(0, 0)$: $m_2 = 0$, but commutator still has $-(N_c - 1)$ contribution from the N_c BST primary

This means the substrate's discreteness signature is charge-modulated but never vanishes (the $N_c - 1 = 2$ factor persists).

3.3 SWPP three-phase cycle in language of $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$

Per SWPP RATIFIED Casey-named: substrate executes absorption $\rightarrow$ commitment $\rightarrow$ emission per Koons tick.

The $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ commutator gives the commitment-phase non-commutativity content: - During absorption (input recording): substrate reads K-type content; $\hat{H}_{\text{sub}}$ acts diagonally (no time evolution yet) - During commitment (substrate work): $\hat{T}_{\text{tick}}$ acts, advancing K-type via the field-arithmetic step; this is where $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ non-vanishing manifests - During emission (output communication): result projected onto Shilov boundary via Bergman kernel

The commitment phase IS the operation that produces $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] \neq 0$ — it's the substrate's irreversible computational step per cycle.

3.4 Connection to Koons tick $t_K = t_P \cdot \alpha^{(C_2^2)}$ (T2405)

T2405 RIGOROUSLY CLOSED: Koons tick $t_K \approx 10^{(-120)} \text{ s} = t_{\text{Planck}} \cdot \alpha^{(C_2^2)}$ where $C_2 = 6$.

The commutator $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] \cdot t_K$ gives the operator displacement per tick: $\Delta \hat{H}_{\text{sub}}$ per tick $\approx \langle V_K | [\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] | V_K \rangle \cdot (\text{something}) = -(2m_2 + N_c - 1) \cdot \text{matrix element}$

For the substrate ground state $V_-(0, 0)$: $\Delta \approx -2$ per tick on the imaginary-time scale t_K .

This is a substrate-natural “minimum action quantum” — the smallest non-trivial substrate operation. The Bell sub-Tsirelson $1/8$ deviation (T2399) and DCCP signature $\Delta_{\text{DCCP}} = 1/N_{\text{max}}$ (Toy 3516) both ultimately trace to this per-tick discreteness quantum.

3.5 Connection to 8-sided die mechanism (A_{sub} v0.4 Section 6)

The $2^{N_c} = 8$ commitment paths per tick may correspond to: - 2 binary choices per Cartan generator $\times N_c = 3$ Cartan generators - 8 sign-choices for $(\hat{Q}, \hat{P}_{\text{op}}, \hat{\gamma}^5)$ eigenvalues (since these are the substrate's discrete Z_2 -like operators)

Per $[\hat{Q}, \hat{P}_{\text{op}}]$ step 1 + $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ step 2: - $\hat{Q}$ and $\hat{P}_{\text{op}}$ anti-commute (step 1) $\rightarrow$ only certain (Q, P) combinations diagonalizable simultaneously - $\hat{T}_{\text{tick}}$ acts non-trivially with $\hat{H}_{\text{sub}}$ (step 2) $\rightarrow$ K-type advancement scales with charge - Combined: 8 potential commitment paths are constrained by (Q, P) simultaneous-diagonalizability + charge-modulated discreteness

This is two-piece evidence for the “1 unstable face” hypothesis — A_sub v0.4 Section 6.3 multi-week resolution work.

4. Honest scope (Cal #27 STANDING)

What’s STRUCTURALLY VERIFIED CANDIDATE in v0.1: - Algebra of $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] = -(2\hat{Q} + N_c - 1) \cdot \hat{T}_{\text{tick}}$ is straightforward given the SIMPLEST-MODEL $\hat{T}_{\text{tick}} : V(m_1, m_2) \rightarrow V(m_1, m_2 + 1)$ - Substrate discreteness signature (Section 3.1) is structurally implied by $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] \neq 0$ - Charge-weighted discreteness pattern (Section 3.2) is direct algebra

What’s MODEL-DEPENDENT in v0.1 (key honest-scope flag): - The simplest model $\hat{T}_{\text{tick}} : V(m_1, m_2) \rightarrow V(m_1, m_2 + 1)$ is a CANDIDATE not the only possibility - Full K59 cyclotomic 7-step mechanism on GF(128) likely refines $\hat{T}_{\text{tick}}$ to a more complex 7-step transition pattern - The numerical result $(2\hat{Q} + N_c - 1)$ factor depends on the simplest-model assumption - Multi-week v0.2 work: derive $\hat{T}_{\text{tick}}$ from K59 mechanism + recompute commutator

What’s INTERPRETIVE in v0.1: - Section 3.3 SWPP commitment-phase non-commutativity reading - Section 3.4 Bell 1/8 + DCCP 1/N_max ultimately tracing to per-tick discreteness quantum - Section 3.5 8-sided die structural constraint reading

Cal #27 STANDING reflexive trigger: 1 trigger this doc (Section 3.3 SWPP three-phase mapping feels substrate-natural; checked — it’s an interpretive elevation of standard SWPP framing, not new derivation; flagged as INTERPRETIVE in v0.1).

Forward-derivation discipline maintained: result computed from $\hat{T}_{\text{tick}} + \hat{H}_{\text{sub}}$ definitions, not back-fit to a target. If the K59 cyclotomic refinement (v0.2 multi-week work) shows the simplest model gives wrong substrate-mechanism predictions, that’s honest negative result and we refine.

5. Next commutator (A_sub v0.5 step 3 — load-bearing)

Per A_sub v0.4 Section 3.3 Section 12 recommended trio: with steps 1 + 2 closed, step 3 is $[\hat{B}, \hat{T}_{\text{tick}}]$ — Bell-CHSH dynamical structure (load-bearing for Track DC Bell 1/8 mechanism).

Recommended next: $[\hat{B}, \hat{T}_{\text{tick}}]$ computation. With $\hat{B} = (C_2/2^{\text{rank}^2}) \cdot |V_{(0,0)}\rangle\langle V_{(0,0)}|$ (T2399 rank-1 projector framework) and $\hat{T}_{\text{tick}}$ advancing K-types per tick, this commutator captures substrate Bell-test dynamics.

6. Graph framing — edge structure from $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$

Per A_sub v0.4 Section 3 graph framing: the $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ commutator gives:

****Edge from $\hat{T}_{\text{tick}}$ **:** advances K-type $V_{(m_1, m_2)} \rightarrow V_{(m_1, m_2 + 1)}$ per tick (S^1 -derived edge per Half-Integer Axis G v0.2 partition Section 3 row 9).

Edge weight from $\hat{H}_{\text{sub}}$: each K-type has self-loop with Casimir eigenvalue $C_2(K)$ as weight.

Composite edge from $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$: K-type advancement edge weighted by $-(2m_2 + N_c - 1)$. Edges carry charge-weighted discreteness signature.

The K-type graph's edges (per A_sub v0.4 Section 3) have weight rule that includes this discreteness factor. Higher-charge K-types have stronger substrate-tick transition weighting in the graph dynamics.

7. Coordination

Cal: cold-read of Section 2.1 algebra (mechanical) + tier-discipline check on Section 3.4 + 3.5 interpretive content; flag any K59 cyclotomic refinement needed.

Keeper: K-audit chain entry for $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] = -(2\hat{Q} + N_c - 1) \cdot \hat{T}_{\text{tick}}$ structural identity (model-dependent).

Elie: K52a Session 7+ multi-year work on substrate Hamiltonian $\hat{H}_{\text{sub}}$ may inform $\hat{T}_{\text{tick}}$ refinement; Toy 3531 K-type population can use this commutator to test substrate dynamics.

Grace: catalog entry for substrate discreteness signature $||[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]|| / ||\hat{H}_{\text{sub}}||$ as candidate observable; cross-reference to T2405 Koons tick + T2441 Casimir vacuum.

— Lyra, A_sub commutator closure step 2 $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}]$ v0.1 filed Tuesday 2026-05-26 ~08:40 EDT per Task #322 v0.4 Section 3.3 work plan. STRUCTURALLY VERIFIED CANDIDATE pending Cal cold-read of Section 2.1 algebra + model-dependency flag for K59 cyclotomic refinement (v0.2 multi-week). Substantive finding: substrate evolution is structurally discrete; $[\hat{T}_{\text{tick}}, \hat{H}_{\text{sub}}] \neq 0$ is the substrate's discreteness signature, scaling with $(2\hat{Q} + N_c - 1)$ — charge-modulated discreteness, with $N_c - 1 = 2$ baseline persisting for neutral K-types.